

Chapter 01 Measurements

Fundamental Quantities and Their Dimension

length(L), mass(m), time(t) and other physical quantities can be constructed from these three.

System of Measurement

SI--International system, also called mks units.

Dimensional Analysis

Ensure the both sides of equation must have the same dimensions

Length

To measure a small length

The minimum of a microscope:

$$\delta = \frac{\lambda}{NA}$$

where NA is numerical aperture, δ is spatial resolution

Uncertainty in Measurements

A kind uncertainty : u_A

The component that can be processed by **statistical methods** when multiple measurements are made with equal precision.

$$u_A = t \cdot \sqrt{\frac{(\sum_{i=1}^n x_i - \bar{x})^2}{n(n-1)}}$$

where t is related to **the number of measurements and confidence probability**.

B kind uncertainty: u_B

Components for a single measurement

The portion produced by **measurer's estimate** u_{B1} , Maximum tolerance of the **instrument** u_{B2} ;

$$u_B = \sqrt{u_{B1}^2 + u_{B2}^2}$$

$$u = \sqrt{u_A^2 + u_B^2}$$